

SIDDHI SINGH

Backend Developer

Node.js • Express.js • MongoDB • PostgreSQL • Redis
India | siddhisingh5@outlook.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Early-career Backend Engineer with hands-on experience across Node.js, Express.js, MongoDB, PostgreSQL, and Redis. I have built secure REST APIs, authentication systems, multi-tenant applications and asynchronous background job processing during 6 months backend internship and independent project work, backed by 213 passing automated tests across 5 projects.

TECHNICAL HIGHLIGHTS

- Built 5 backend systems spanning APIs, queues, gateways, and webhooks each backed by automated test suite.
- Implemented JWT and Goggle OAuth authentication across multiple applications.
- Designed multi-tenant architecture with role – based access control (RBAC).
- Developed asynchronous background processing using BullMQ and Redis, including retry and dead-letter-queue handling.
- Built a centralized API gateway with Redis-based rate limiting and a circuit breaker pattern.
- Secured service-to-service and webhook communication with HMAC signature verification.

TECHNICAL SKILLS

Language: JavaScript, SQL

Backend: Node.js, Express.js, REST APIs, Socket.io

Database: MongoDB, PostgreSQL, Redis (Upstash, Docker)

Architecture: RBAC, Multi-tenancy, API Gateway, BullMQ, Circuit Breaker

Security: JWT, OAuth 2.0, Passport.js, Helmet, Bcrypt, Rate Limiting, HMAC Verification

Testing: Vitest, Supertest, mongodb-memory-server

Developer Tool: Docker, Git, GitHub, Swagger, Postman

API: Gemini API, Resend, ImageKit

PROFESSIONAL EXPERIENCE

Web Developer Intern (Backend) | Himritva

Project: Projectify – multi-tenant task/project management platform with real-time collaboration – [GitHub](#) | (Node.js, Express, MongoDB, Redis, Socket.io) --

- JWT Authentication – short-lived access/refresh tokens (15 min/7 days) in HttpOnly cookies, reducing client-side JS exposure to auth tokens, Google OAuth as secondary login.
- Workspace RBAC – Owner/Admin/Member roles enforced in middleware, keeping permission logic out of controllers.
- Multi-Tenant Data Isolation – Membership checks plus workspaceId-filtered queries on every request restrict access to a user's own workspace data.
- Task Management APIs – CRUD with fractional-position kanban ordering, so drag-reorder writes one document, not the whole board.
- Real-Time Task Updates – Socket.io project scoped rooms deliver task events only to users viewing that board.
- Testing – 31 tests (Vitest, Supertest): 11 Auth, 7 RBAC, 7 Tenant-isolation, 6 Socket.io. 31/31 passing.

Additional Experience

Freelance Consulting (Design & Client Delivery) 2021-2025 – alongside BCA and MCA studies, 6+clients engagements covering requirement gathering, delivery and stakeholder communication.

BACKEND ENGINEERING PROJECTS

API Gateway

Centralized API gateway for authentication, traffic control and service routing – [GitHub](#) / Tech Stack: Node.js, Express, Redis

- Built an API gateway centralizing JWT + API-key auth, rate limiting and upstream failure protection in one pipeline.
- Implemented Redis sorted-set sliding-window rate limiting to close the fixed-window boundary-burst bypass.
- Designed a closed/open/half-open circuit breaker (3 failures, 30s cooldown) with 8secs upstream timeout.
- Secured API-key checks with crypto.timingSafeEqual to block timing attacks, stripped auth headers before proxying.
- Wrote 94 tests (Vitest, Supertest) across 5 files covering auth, rate limits, circuit-breaker states and proxying. 94/94 passing.

Email Job Queue

Asynchronous background job processing service for reliable email delivery – [GitHub](#) | Tech Stack: Node.js, Express, BullMQ, Redis, PostgreSQL

- Built an async email pipeline (Express+BullMQ+Redis) that queue the job and returns 202 immediately, decoupling delivery from the request.
- Implemented priority queuing (1-10) so password-reset emails jump ahead of newsletters, with 5-way worker concurrency.
- Added exponential-backoff retries (5s -> 25s ->125s) with a dead-letter queue to handle failed delivery attempts after 3 failures.
- Designed a PostgreSQL layer that writes the job row before queuing (no orphaned jobs), tracking status/attempts/errors.
- Wrote 27 tests (16-unit, 11 integration Vitest) covering the email service, worker retry/DLQ logic and job-status APIs. 27/27 passing.

Webhook Delivery Service

Webhook delivery platform for reliable event distribution to third-party integrations – [GitHub](#) | Tech Stack: Node.js, Express, BullMQ, Redis, MongoDB

- Built a webhook platform (Express+BullMQ+Redis+MongoDB) queuing one job per subscriber per event – isolating failures.
- Secured payloads with HMAC-SHA256 signing and a 5-min timestamp window that rejects stale signed requests.
- Implemented 5 stage exponential backoff (5s->3125s) per subscriber, persisting full delivery history in MongoDB for audit.
- Wrote 35 tests (24-unit, 11 integration Vitest) covering signature verification, replay protection, and retries. 35/35 passing.

AI Interview Prep

AI-assisted interview preparation platform with personalized question generation – [GitHub](#) | Tech Stack: Node.js, Express, Gemini API, MongoDB

- Built an AI interview-prep platform using the Gemini API for personalized questions with session management and auth.
- Wrote 26 tests (11 session, 15 question-gen Vitest) covering Gemini mocking, malformed output and authorization. 26/26 passing.

EDUCATION

Master of Computer Applications (MCA) – SNDT Women's University, 2025 – CGPA 8.13

Bachelor of Computer Applications (BCA) – SNDT Women's University, 2022 – CGPA 9.11

LEADERSHIP & ACHIEVEMENTS

Placement Head (MCA) • President, Infomaniac IT Club • Department Best Designer • College Fest Lead Organizer & Anchor